COMPOSITION OF INTERNAL HOM ADMITS A CANONICAL FUNCTORIAL MORPHISM IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'We omit the proof that the construction is functorial.' at the end of the counted proof span; that sentence is kept verbatim and marks the step left to the reader. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The composition map for Hom complexes of K-injective representatives. invoked at proof lines 18771-18778. Is the underlying map of complexes whose lift to the derived category is constructed.
- K-flat replacements and the fact that a K-flat complex computes the derived tensor product. used at proof lines 18782-18793. Is what makes the source of the first displayed arrow represent the derived tensor product.
- Hom complexes of K-injective representatives representing the derived Hom. used at proof lines 18779-18781. Identifies the three Hom complexes with the objects named in the statement.

Source reference. Original lemma 17966: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17966–17966. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Given K, L, M in $D(R)$ there is a canonical morphism*

$$R\operatorname{Hom}_R(L, M) \otimes^{\mathbf{L}}_R R\operatorname{Hom}_R(K, L) \longrightarrow R\operatorname{Hom}_R(K, M)$$

in $D(R)$ functorial in K, L, M.

Proof. Choose a K-injective complex $I^\bullet$ representing M, a K-injective complex $J^\bullet$ representing L, and any complex of R-modules $K^\bullet$ representing K. By Lemma 17966 there is a map of complexes

$$\operatorname{Tot}\left(\operatorname{Hom}^\bullet(J^\bullet, I^\bullet) \otimes_R \operatorname{Hom}^\bullet(K^\bullet, J^\bullet)\right) \longrightarrow \operatorname{Hom}^\bullet(K^\bullet, I^\bullet)$$

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

The complexes of R-modules $\mathrm{Hom}^\bullet(J^\bullet, I^\bullet)$, $\mathrm{Hom}^\bullet(K^\bullet, J^\bullet)$, and $\mathrm{Hom}^\bullet(K^\bullet, I^\bullet)$ represent $R\mathrm{Hom}_R(L, M)$, $R\mathrm{Hom}_R(K, L)$, and $R\mathrm{Hom}_R(K, M)$. If we choose a K-flat complex $H^\bullet$ and a quasi-isomorphism $H^\bullet \to \mathrm{Hom}^\bullet(K^\bullet, J^\bullet)$, then there is a map

$$\mathrm{Tot}\left(\mathrm{Hom}^\bullet(J^\bullet, I^\bullet) \otimes_R H^\bullet\right) \longrightarrow \mathrm{Tot}\left(\mathrm{Hom}^\bullet(J^\bullet, I^\bullet) \otimes_R \mathrm{Hom}^\bullet(K^\bullet, J^\bullet)\right)$$

whose source represents $R\mathrm{Hom}_R(L, M) \otimes^{\mathbf{L}}_R R\mathrm{Hom}_R(K, L)$. Composing the two displayed arrows gives the desired map. We omit the proof that the construction is functorial. $\square$